

Capstone code Report

File structure

TestAutomationPractice

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Main Test

```
package com.test.automation;

import org.testng.annotations.Test;
import POM.POM_Implementation;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.DataProvider;
import java.io.FileInputStream;
import java.io.IOException;
import java.net.MalformedURLException;
import java.net.URL;
import org.apache.poi.xssf.usermodel.XSSFSheet;
import org.apache.poi.xssf.usermodel.XSSFWorkbook;
import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.testng.annotations.AfterMethod;

public class BaseTest {

    WebDriver driver;

    WebDriverWait wait;

    JavascriptExecutor js;

    POM_Implementation p;

    //@DataProvider(name="NameExcel")
```

```

//public Object[][] nameexcel() throws IOException {
//
//    FileInputStream fs = new FileInputStream(
//        "C:\\Users\\surla\\OneDrive\\phone
img\\OneDrive\\Documents\\sample_pdf.xlsx");
//
//    XSSFWorkbook workbook = new XSSFWorkbook(fs);
//    XSSFSheet sheet = workbook.getSheet("sheet1");
//
//    int rows = sheet.getPhysicalNumberOfRows();
//
//    Object[][] data = new Object[rows - 1][1];
//
//    for (int i = 1; i < rows; i++) {
//        data[i - 1][0] = sheet.getRow(i).getCell(0).toString();
//        System.out.println("Name = " + data[i - 1][0]);
//    }
//
//    workbook.close();
//    fs.close();
//
//    return data;
//}
//@DataProvider(name="EmailExcel")
//public Object[][] emailexcel() throws IOException{
//FileInputStream fs=new

```

```

//FileInputStream("C:\\Users\\surla\\OneDrive\\phone
img\\OneDrive\\Documents\\sample.pdf.xlsx");

//XSSFWorkbook workbook=new XSSFWorkbook(fs);

//XSSFSheet sheet=workbook.getSheet("sheet1");

//int rows=sheet.getPhysicalNumberOfRows();

//int cols=sheet.getRow(0).getLastCellNum();

//Object[][] data=new Object[rows-1][cols];

//for(int i=1;i<rows;i++) { for(int j=0;j<cols;j++) {

//data[i-1][j]=sheet.getRow(i).getCell(1).toString();

//}

//}

//fs.close();

//workbook.close();

//return data;

//}

//@DataProvider(name="MobileNumberExcel")

//public Object[][] mobilenumberexcel() throws IOException{

//FileInputStream fs=new

//FileInputStream("C:\\Users\\surla\\OneDrive\\phone
img\\OneDrive\\Documents\\sample.pdf.xlsx");

//XSSFWorkbook workbook=new XSSFWorkbook(fs);

//XSSFSheet sheet=workbook.getSheet("sheet1");

//int rows=sheet.getPhysicalNumberOfRows();

//int cols=sheet.getRow(0).getLastCellNum();

//Object[][] data=new Object[rows-1][cols];

//for(int i=1;i<rows;i++) {

//for(int j=0;j<cols;j++) {

```

```

//data[i-1][j]=sheet.getRow(i).getCell(2).toString();
//}
//} fs.close();
//workbook.close();
//return data;
//}
//@DataProvider(name="AddressExcel")
//public Object[][] addressexcel() throws IOException{
//FileInputStream fs=new
//FileInputStream("C:\\Users\\surla\\OneDrive\\phone
img\\OneDrive\\Documents\\sample.pdf.xlsx");
//XSSFWorkbook workbook=new XSSFWorkbook(fs);
//XSSFSheet sheet=workbook.getSheet("sheet1");
//int rows=sheet.getPhysicalNumberOfRows();
//int cols=sheet.getRow(0).getLastCellNum();
//Object[][] data=new Object[rows-1][cols];
//for(int i=1;i<rows;i++) {
//for(int j=0;j<cols;j++) {
//data[i-1][j]=sheet.getRow(i).getCell(3).toString();
//}
//}
//fs.close();
//workbook.close();
//return data;
//}
@DataProvider(name="searchdata")
public Object[][] data(){

```

```

return new Object[][] {
{"King"}, };
}

@DataProvider(name="paragraph")
public Object[][] paragraphbox(){
return new Object[][] {
{"You know why my World is Beautiful?", "Why??", "Because, it's you!."
}};
}

@BeforeMethod
public void beforeMethod() throws InterruptedException,
MalformedURLException {
driver=new ChromeDriver();
p=new POM_Implementation(driver);
driver.get("https://testautomationpractice.blogspot.com/");
Thread.sleep(3000);
driver.manage().window().maximize();
}

@Test
public void fillForm() throws InterruptedException {

    p.Name_Element("Bala");
    p.Email_Element("bala@gmail.com");
    p.Ponenumber_Element("9547783210");
    p.Address("Vizag");
}

```

```

@Test(dataProvider="NameExcel",priority=1) public void
GUI_ELEMENTSName(String name) throws
//InterruptedException {
//p.Name_Element(name);
//}
@Test(dataProvider="EmailExcel",priority=2)
//public void GUI_ELEMENTSEmail(String email) throws
//InterruptedException {
//p.Email_Element(email);
//}
@Test(dataProvider="MobileNumberExcel",priority=3)
//public void GUI_ELEMENTSMobileNumber(String mobilenummer)
//throws InterruptedException {
//p.Ponenumber_Element(mobilenummer);
//}
@Test(dataProvider="AddressExcel",priority=4)
//public void GUI_ELEMENTSAAddress(String address) throws
//InterruptedException {
//p.Address(address);
//}
@Test(priority=5)
public void Radio_and_CheckButtons() throws InterruptedException {
p.radio_checkbox();
}
@Test(priority=6) public void uploadfiles() throws InterruptedException {
p.upload();
}

```



```

@Test(priority=7)
public void Pagination_Table() throws InterruptedException {
    p.page_table();
}

@Test(dataProvider="paragraph",priority=8)
public void Forms(String firstparagraph,String secondparagraph,String
thirdparagraph) throws InterruptedException {
    p.forms(firstparagraph,secondparagraph,thirdparagraph);
}

@Test(dataProvider="searchdata",priority=9)
public void Search(String searchitem) throws InterruptedException {
    p.search(searchitem);
}

@Test(priority=10)
public void Dynamicbutton() throws InterruptedException {
    p.dynamicbtn();
}

@Test(priority=11)
public void AlertsandPopups() throws InterruptedException {
    p.alertaccept(); }

@Test(priority=12)
public void window_Handling() throws InterruptedException {
    p.newtab();
}

@Test(priority=13)
public void multiple_windowHandling() throws InterruptedException {

```

```
p.multiplewindow();  
}  
@Test(priority=14)  
public void Mouse_Hover() throws InterruptedException {  
    p.movetoelement();  
}  
@Test(priority=15)  
public void DoubleClick() throws InterruptedException {  
    p.doubleclick();  
}  
@Test(priority=16)  
public void DragandDrop() throws InterruptedException {  
    p.draganddrop();  
}  
@Test(priority=17) public void Slider() throws InterruptedException {  
    p.slider();  
}  
@Test(priority=18)  
public void Scrolling_Dropdown() throws InterruptedException {  
    p.scrolling_dropdown();  
}  
@Test(priority=19)  
public void LaptopLinks() throws InterruptedException {  
    p.links();  
}  
//@Test(priority=20)
```

```
//public void Broken_Links() throws InterruptedException, IOException {  
//p.broken_links();  
//}  
  
@AfterMethod  
public void afterMethod() {  
driver.quit();  
}  
}
```

POM_Implementation Class

```
package POM;  
  
import java.io.File;  
import java.io.IOException;  
import java.time.Duration;  
import java.util.Set;  
  
import org.apache.commons.io.FileUtils;  
import org.openqa.selenium.Alert;  
import org.openqa.selenium.By;  
import org.openqa.selenium.JavascriptExecutor;  
import org.openqa.selenium.Keys;  
import org.openqa.selenium.OutputType;  
import org.openqa.selenium.TakesScreenshot;  
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.Select;
import org.openqa.selenium.support.ui.WebDriverWait;
import org.testng.Assert;
```

```
public class POM_Implementation {
```

```
    WebDriver driver;
```

```
    JavascriptExecutor js;
```

```
    WebDriverWait wait;
```

```
    TakesScreenshot ts;
```

```
    public POM_Implementation(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    By namefield = By.id("name");
```

```
    By emailfield = By.id("email");
```

```
    By phonefield = By.id("phone");
```

```
    By addressfield = By.id("textarea");
```

```
    By radiobutton = By.id("male");
```

```
    By sundaycheckbox = By.id("sunday");
```

```
By mondaycheckbox = By.id("monday");
By tuesdaycheckbox = By.id("tuesday");
By wednesdaycheckbox = By.id("wednesday");
By thursdaycheckbox = By.id("thursday");
By fridaycheckbox = By.id("friday");
By saturdaycheckbox = By.id("saturday");
By countrydropdown = By.id("country");
By colorselect = By.id("colors");
By Listselect = By.id("animals");
By datepicker1 = By.id("datepicker");
By datepicker2 = By.id("txtDate");
By datepicker2year = By.className("ui-datepicker-year");
By datepicker2month = By.className("ui-datepicker-month");
By datepicker3startdate = By.id("start-date");
By datepicker3enddate = By.id("end-date");
By submitbutton = By.className("submit-btn");
```

```
// Files
```

```
By singlefileupload = By.id("singleFileInput");
By multiplefileupload = By.id("multipleFilesInput");
```

```
// Pagination Table
```

```
By checkbox1 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[1]/td[4]/input");

By checkbox2 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[2]/td[4]/input");
```

By checkbox3 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[3]/td[4]/input");

By checkbox4 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[4]/td[4]/input");

By checkbox5 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[5]/td[4]/input");

By secondpage = By.linkText("2");

By checkbox6 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[1]/td[4]/input");

By checkbox7 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[2]/td[4]/input");

By checkbox8 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[3]/td[4]/input");

By checkbox9 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[4]/td[4]/input");

By checkbox10 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[5]/td[4]/input");

By thirdpage = By.linkText("3");

By checkbox11 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[1]/td[4]/input");

By checkbox12 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[2]/td[4]/input");

By checkbox13 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[3]/td[4]/input");

By checkbox14 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[4]/td[4]/input");

By checkbox15 =
By.xpath("//*[@id=\"productTable\"]/tbody/tr[5]/td[4]/input");

By fourthpage = By.linkText("4");

```
By checkbox16 =  
By.xpath("//*[@id=\"productTable\"]/tbody/tr[1]/td[4]/input");
```

```
By checkbox17 =  
By.xpath("//*[@id=\"productTable\"]/tbody/tr[2]/td[4]/input");
```

```
By checkbox18 =  
By.xpath("//*[@id=\"productTable\"]/tbody/tr[3]/td[4]/input");
```

```
By checkbox19 =  
By.xpath("//*[@id=\"productTable\"]/tbody/tr[4]/td[4]/input");
```

```
By checkbox20 =  
By.xpath("//*[@id=\"productTable\"]/tbody/tr[5]/td[4]/input");
```

```
// Forms
```

```
By firsttextbox = By.id("input1");
```

```
By secondtextbox = By.id("input2");
```

```
By thirdtextbox = By.id("input3");
```

```
By submitbutton1 = By.id("btn1");
```

```
By submitbutton2 = By.id("btn2");
```

```
By submitbutton3 = By.id("btn3");
```

```
// Dynamic button
```

```
By dynamicbuttonclick = By.name("start");
```

```
// new tab button
```

```
By newtabclickbutton = By.xpath("//*[@id=\"HTML4\"]/div[1]/button");
```

```
By popupwindow = By.id("PopUp");
```

```
// mouse
```

```
By mousehover = By.xpath("//*[@id=\"HTML3\"]/div[1]/div/button");
```

```
// doubleClick copy
```

```
By copybutton = By.xpath("//button[@ondblclick=\"myFunction1()\"]");
```

```
// Drag and Drop
```

```
By source = By.id("draggable");
```

```
By destination = By.id("droppable");
```

```
// Slider
```

```
By pointer1 = By.xpath("//*[@id=\"slider-range\"]/span[1]");
```

```
By pointer2 = By.xpath("//*[@id=\"slider-range\"]/span[2]");
```

```
// ScrollDown
```

```
By Dropdown = By.id("comboBox");
```

```
By item = By.xpath("//*[@id=\"dropdown\"]/div[5]");
```

```
// Search
```

```
By searchbox = By.id("Wikipedia1_wikipedia-search-input");
```

```
By searchbutton = By.xpath("//*[@id=\"Wikipedia1_wikipedia-search-form\"]/div/span[2]/span[2]/input");
```

```
By searchsuggestion = By.linkText("King");
```

```
// Links
```

```
By applelink = By.id("apple");
```

```
By lenovolink = By.id("lenovo");
```

```
By delllink = By.id("dell");
```



```
// AlertButton
```

```
By alertbutton = By.id("alertBtn");
```

```
// BrokenLinks
```

```
By Errorcode400 = By.linkText("Errorcode 400");
```

```
By Errorcode401 = By.linkText("Errorcode 401");
```

```
By Errorcode403 = By.linkText("Errorcode 403");
```

```
By Errorcode404 = By.linkText("Errorcode 404");
```

```
By Errorcode408 = By.linkText("Errorcode 408");
```

```
By Errorcode500 = By.linkText("Errorcode 500");
```

```
By Errorcode502 = By.linkText("Errorcode 502");
```

```
By Errorcode503 = By.linkText("Errorcode 503");
```

```
public void Name_Element(String name) throws InterruptedException {
```

```
    js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,100)");
```

```
    Thread.sleep(400);
```

```
    wait = new WebDriverWait(driver, Duration.ofSeconds(10));
```

```
    WebElement firstname =
```

```
wait.until(ExpectedConditions.elementToBeClickable(namefield));
```

```
firstname.click();
```

```
Thread.sleep(400);
```

```
System.out.println("name = " + name);
```

```
firstname.sendKeys(name);
```

```
Thread.sleep(400);
```

```
System.out.println("Firstname entered successfully");
```

```
}
```

```
public void Email_Element(String email) throws InterruptedException {
```

```
js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,150)");
```

```
Thread.sleep(400);
```

```
WebElement Email = driver.findElement(emailfield);
```

```
wait = new WebDriverWait(driver, Duration.ofSeconds(10));
```

```
wait.until(ExpectedConditions.elementToBeClickable(emailfield));
```

```
Email.click();
```

```
Thread.sleep(400);
```

```
System.out.println("email = " + email);
```

```
Email.sendKeys(email);
```

```
Thread.sleep(400);
```

```
System.out.println("Email entered successfully");
```

```
}
```

```
public void Phonenumber_Element(String mobilenum) throws  
InterruptedException {
```

```
js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,200)");
```

```
Thread.sleep(400);
```

```
WebElement phonenumber = driver.findElement(phonefield);
```

```
phonenumber.click();
```

```
Thread.sleep(400);
```

```
    phonenumber.sendKeys(mobilenumber);
    System.out.println("mobilenumber = " + mobilenumber);

    Thread.sleep(400);

    System.out.println("Mobilenumber entered successfully");
}

public void Address(String address) throws InterruptedException {

    js = (JavascriptExecutor) driver;

    js.executeScript("window.scrollTo(0,200)");

    Thread.sleep(400);

    WebElement addres = driver.findElement(addressfield);

    addres.click();

    Thread.sleep(400);

    addres.sendKeys(address);
    Thread.sleep(400);
```

```
System.out.println("address = " + address);
```

```
System.out.println("Address entered successfully");
```

```
Thread.sleep(400);
```

```
}
```

```
public void radio_checkbox() throws InterruptedException {
```

```
    js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,900)");
```

```
    driver.findElement(radiobutton).click();
```

```
    Thread.sleep(400);
```

```
    driver.findElement(sundaycheckbox).click();
```

```
//
```

```
// Thread.sleep(400);
```

```
//
```

```
// driver.findElement(mondaycheckbox).click();
```

```
//
```

```
// Thread.sleep(400);
```

```
//
```

```
//      driver.findElement(tuesdaycheckbox).click();
//
//      Thread.sleep(400);
//
//      driver.findElement(wednesdaycheckbox).click();
//
//      Thread.sleep(400);
//
//      driver.findElement(thursdaycheckbox).click();
//
//      Thread.sleep(400);
//
//      driver.findElement(fridaycheckbox).click();

      Thread.sleep(400);

      driver.findElement(saturdaycheckbox).click();

      System.out.println("Weekday checkbox clicked");

      Thread.sleep(400);

      WebElement selectcountry =
driver.findElement(countrydropdown);

      selectcountry.click();
```

```
Thread.sleep(400);
```

```
Select s = new Select(selectcountry);
```

```
s.selectByVisibleText("India");
```

```
System.out.println("Country Selected successfully");
```

```
Thread.sleep(400);
```

```
WebElement selectcolor = driver.findElement(colorselect);
```

```
selectcolor.click();
```

```
Thread.sleep(400);
```

```
Select color = new Select(selectcolor);
```

```
color.selectByVisibleText("Blue");
```

```
System.out.println("Colors Selected successfully");
```

```
WebElement selectsortedList = driver.findElement(Listselect);
```

```
selectsortedList.click();
```

```
Thread.sleep(400);
```

```
Select sortedlist = new Select(selectsortedList);
```

```
sortedlist.selectByVisibleText("Lion");
```

```
System.out.println("Animals Selected successfully");
```

```
Thread.sleep(400);
```

```
js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,300)");
```

```
Thread.sleep(400);
```

```
WebElement firstdate = driver.findElement(datepicker1);
```

```
Thread.sleep(400);
```

```
firstdate.sendKeys("07/16/2026");
```

```
Thread.sleep(400);
```

```
System.out.println("Firstdate entered successfully");
```



```
driver.findElement(datepicker1).sendKeys(Keys.ENTER);
```

```
WebElement seconddate=driver.findElement(datepicker2);
```

```
seconddate.click();
```

```
Thread.sleep(400);
```

```
WebElement year=driver.findElement(datepicker2year);
```

```
Select y=new Select(year);
```

```
y.selectByVisibleText("2026");
```

```
Thread.sleep(400);
```

```
WebElement month=driver.findElement(datepicker2month);
```

```
Select m=new Select(month);
```

```
m.selectByVisibleText("May");
```

```
Thread.sleep(400);
```

```
driver.findElement(By.xpath("//a[text()='17']")).click();
```

```
System.out.println("Second date entered successfully");
```

```
Thread.sleep(400);
```

```
WebElement thirdstartdate =  
driver.findElement(datepicker3startdate);
```

```
thirdstartdate.sendKeys("12-06-2026");
```

```
Thread.sleep(400);
```

```
WebElement thirddenddate =  
driver.findElement(datepicker3enddate);
```

```
thirddenddate.sendKeys("5-6-2027");
```

```
System.out.println("Thirddate entered successfully");
```

```
Thread.sleep(400);
```

```
driver.findElement(submitbutton).click();
```

```
System.out.println("Submit button successfully");
```

```
Thread.sleep(400);
```

```
}
```

```

public void upload() throws InterruptedException {

    js = (JavascriptExecutor) driver;

    js.executeScript("window.scrollTo(0,1700)");

    Thread.sleep(400);

    WebElement singlefile = driver.findElement(singlefileupload);

    singlefile.sendKeys(

        "C:\\Users\\surla\\OneDrive\\Desktop\\huskey\\WhatsApp Image 2026-01-07 at 11.12.41 PM (1).jpeg");

    System.out.println("Single File Uploaded Successfully");

    Thread.sleep(400);

    WebElement multiplefiles =
driver.findElement(multiplefileupload);

    multiplefiles.sendKeys(

        "C:\\Users\\surla\\OneDrive\\Desktop\\huskey\\WhatsApp Image 2026-01-07 at 11.12.41 PM (1).jpeg" + "\n"

```

+
"C:\\Users\\surla\\OneDrive\\Desktop\\huskey\\WhatsApp Image 2026-01-07
at 11.12.42 PM (1).jpeg");

System.**out**.println("Multiple File Uploaded Successfully");

Thread.*sleep*(400);

}

public void page_table() throws InterruptedException {

js = (JavascriptExecutor) driver;

js.executeScript("window.scrollTo(0,2700)");

Thread.*sleep*(400);

driver.findElement(checkbox1).click();

Thread.*sleep*(400);

driver.findElement(checkbox2).click();

Thread.*sleep*(400);

driver.findElement(secondpage).click();

```
Thread.sleep(400);
```

```
driver.findElement(checkbox6).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(checkbox7).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(checkbox8).click();
```

```
Thread.sleep(400);
```

```
//
```

```
// driver.findElement(checkbox9).click();
```

```
//
```

```
// Thread.sleep(400);
```

```
//
```

```
// driver.findElement(checkbox10).click();
```

```
//
```

```
// Thread.sleep(400);
```

```
driver.findElement(thirdpage).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(checkbox15).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(fourthpage).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(checkbox17).click();
```

```
Thread.sleep(400);
```

```
driver.findElement(checkbox20).click();
```

```
Thread.sleep(400);
```

```
}
```

```
public void forms(String firstparagraph, String secondparagraph, String  
thirdparagraph)
```

```
throws InterruptedException {
```

```
js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,100)");
```

```
Thread.sleep(400);
```

```
driver.findElement(firsttextbox).sendKeys(firstparagraph);
```

```
Thread.sleep(400);
```

```
driver.findElement(submitbutton1).click();
```

```
System.out.println("Form 1 Submitted Successfully");
```

```
Thread.sleep(400);
```

```
driver.findElement(secondtextbox).sendKeys(secondparagraph);
```

```
Thread.sleep(400);
```

```
driver.findElement(submitbutton2).click();
```

```
System.out.println("Form 2 Submitted Successfully");
```

```
Thread.sleep(400);
```

```
driver.findElement(thirdtextbox).sendKeys(thirdparagraph);
```

```
Thread.sleep(400);
```

```
driver.findElement(submitbutton3).click();
```

```
System.out.println("Form 3 Submitted Successfully");
```

```
Thread.sleep(400);  
}
```

```
public void search(String searchitem) throws InterruptedException {
```

```
    WebElement text = driver.findElement(searchbox);
```

```
    text.click();
```

```
    Thread.sleep(400);
```

```
    System.out.println("Search textbox Clicked Successfully");
```

```
    text.sendKeys(searchitem);
```

```
    Thread.sleep(400);
```

```
    driver.findElement(searchbutton).click();
```

```
    System.out.println("Item Searched Successfully");
```

```
    Thread.sleep(1000);
```

```
    String pwindow = driver.getWindowHandle();
```



```
driver.findElement(searchsuggestion).click();
```

```
Thread.sleep(400);
```

```
Set<String> nwindow = driver.getWindowHandles();
```

```
for (String window : nwindow) {
```

```
    if (!window.equals(pwindow)) {
```

```
        driver.switchTo().window(window);
```

```
    }
```

```
}
```

```
Thread.sleep(400);
```

```
driver.close();
```

```
driver.switchTo().window(pwindow);
```

```
}
```

```
public void dynamicbtn() throws InterruptedException {
```

```
    js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,900)");
```

```
        driver.findElement(dynamicbuttonclick).click();  
        ;  
  
        System.out.println("Dynamic Start Button Successfully");  
  
        Thread.sleep(400);  
  
    }  
  
    public void alertaccept() throws InterruptedException {  
  
        driver.findElement(alertbutton).click();  
  
        Alert al = driver.switchTo().alert();  
  
        Thread.sleep(400);  
  
        al.accept();  
  
        System.out.println("Simple alert Accepted");  
  
        driver.findElement(By.id("confirmBtn")).click();  
  
        Thread.sleep(400);  
    }  
}
```

```
Alert a = driver.switchTo().alert();
```

```
Thread.sleep(400);
```

```
a.dismiss();
```

```
System.out.println("Confirmation alert dismissed");
```

```
driver.findElement(By.id("promptBtn")).click();
```

```
Alert s = driver.switchTo().alert();
```

```
Thread.sleep(400);
```

```
s.accept();
```

```
System.out.println("Prompt alert Accepted");
```

```
Thread.sleep(400);
```

```
}
```

```
public void newtab() throws InterruptedException {
```

```
js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,100)");
```

```
String oldwindow = driver.getWindowHandle();
```

```
driver.findElement(newtabclickbutton).click();
```

```
Thread.sleep(400);
```

```
Set<String> newwindow = driver.getWindowHandles();
```

```
for (String window : newwindow) {
```

```
    if (!window.equals(oldwindow)) {
```

```
        driver.switchTo().window(window);
```

```
    }
```

```
}
```

```
driver.close();
```

```
driver.switchTo().window(oldwindow);
```

```
Thread.sleep(400);
```

```
System.out.println("Window successfully Handled");
```

```
}
```

```
public void multiplewindow() throws InterruptedException {
```

```
    js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,600)");
```

```
    Thread.sleep(400);
```

```
    String old = driver.getWindowHandle();
```

```
    driver.findElement(popupwindow).click();
```

```
    Thread.sleep(1000);
```

```
    Set<String> newwindows = driver.getWindowHandles();
```

```
    for (String windows : newwindows) {
```

```
        if (!windows.equals(old)) {
```

```
            driver.switchTo().window(windows);
```

```
        }
```

```
    }
```

```
    driver.close();
```

```
        driver.switchTo().window(old);  
    }
```

```
public void movetoelement() throws InterruptedException {
```

```
    js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,700)");
```

```
    WebElement moveto = driver.findElement(mousehover);
```

```
    Actions a = new Actions(driver);
```

```
    a.moveToElement(moveto).perform();
```

```
    System.out.println("Moving pointer to the element is successful");
```

```
    Thread.sleep(400);
```

```
}
```

```
public void doubleclick() throws InterruptedException {
```

```
    JavascriptExecutor js = (JavascriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,1100)");
```

```
Thread.sleep(400);
```

```
WebElement copy = driver.findElement(copybutton);
```

```
Actions c = new Actions(driver);
```

```
c.doubleClick(copy).perform();
```

```
System.out.println("Text copied and pasted");
```

```
Thread.sleep(400);
```

```
}
```

```
public void draganddrop() throws InterruptedException {
```

```
JavaScriptExecutor js = (JavaScriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,1400)");
```

```
Thread.sleep(400);
```

```
WebElement drag = driver.findElement(source);
```

```
WebElement drop = driver.findElement(destination);
```

```
Actions d = new Actions(driver);
```

```
d.dragAndDrop(drag, drop).perform();
```

```
System.out.println("Element dragged and dropped into the  
destination");
```

```
Thread.sleep(400);
```

```
}
```

```
public void slider() throws InterruptedException {
```

```
JavaScriptExecutor js = (JavaScriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,1600)");
```

```
Thread.sleep(400);
```

```
WebElement point1 = driver.findElement(pointer1);
```

```
Actions e = new Actions(driver);
```

```
e.dragAndDropBy(point1, 35, 1).perform();
```

```
System.out.println("Point 1 Slided Successfully");
```



```
Thread.sleep(400);
```

```
WebElement point2 = driver.findElement(pointer2);
```

```
Actions f = new Actions(driver);
```

```
f.dragAndDropBy(point2, 100, 0).perform();
```

```
System.out.println("Point 2 Slided Successfully");
```

```
Thread.sleep(400);
```

```
}
```

```
public void scrolling_dropdown() throws InterruptedException {
```

```
JavaScriptExecutor js = (JavaScriptExecutor) driver;
```

```
js.executeScript("window.scrollTo(0,2100)");
```

```
Thread.sleep(400);
```

```
WebElement drop = driver.findElement(Dropdown);
```

```
drop.click();
```

```
Thread.sleep(400);
```

```
WebElement items = driver.findElement(item);
```

```
items.click();
```

```
Thread.sleep(400);
```

```
}
```

```
public void links() throws InterruptedException {
```

```
    JavascriptExecutor js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,2600)");
```

```
    Thread.sleep(400);
```

```
//
```

```
//     driver.findElement(applelink).click();
```

```
//
```

```
//     Thread.sleep(400);
```

```
//     String expectedtitle = "Apple";
```

```
//
```

```
//     String actualtitle = driver.getTitle();
```

```
//
```

```
//      Assert.assertEquals(expectedtitle, actualtitle, "Title Validation
Failed");
//
//      driver.navigate().back();
//
//      Thread.sleep(400);
//

      driver.findElement(lenovolink).click();

      Thread.sleep(400);

      String expectedurl =
"https://www.lenovo.com/in/en/?orgRef=https%253A%252F%252Ftestautoma
tionpractice.blogspot.com%252F";

      String currenturl = driver.getCurrentUrl();

      if (expectedurl.equals(currenturl)) {

          System.out.println("URL Validation Passed");
      } else {

          System.out.println("URL Validation Failed");
      }

      driver.navigate().back();
```

```
Thread.sleep(400);
```

```
driver.findElement(dellink).click();
```

```
Thread.sleep(400);
```

```
String sourcecode = driver.getPageSource();
```

```
System.out.println("Page source is:" + sourcecode);
```

```
Thread.sleep(400);
```

```
}
```

```
public void broken_links() throws InterruptedException, IOException {
```

```
    JavascriptExecutor js = (JavascriptExecutor) driver;
```

```
    js.executeScript("window.scrollTo(0,2700)");
```

```
    Thread.sleep(400);
```

```
    WebElement title1 = driver.findElement(Errorcode400);
```

```
    title1.click();
```

```
    ts = (TakesScreenshot) driver;
```

```
File src = ts.getScreenshotAs(OutputType.FILE);
```

```
File dest = new File("Screenshot/Error400.png");
```

```
FileUtils.copyFile(src, dest);
```

```
System.out.println("Screenshot of Error 400 took Successfully");
```

```
Thread.sleep(400);
```

```
driver.navigate().back();
```

```
Thread.sleep(400);
```

```
WebElement title2 = driver.findElement(Errorcode401);
```

```
title2.click();
```

```
String text2 = driver.getTitle();
```

```
System.out.println("Error status is" + text2);
```

```
Thread.sleep(400);
```

```
driver.navigate().back();
```

```
Thread.sleep(400);
```

```
//      WebElement title3 = driver.findElement(Errorcode403);  
//  
//      title3.click();  
//  
//      String text3 = driver.getTitle();  
//  
//      System.out.println("Error status is" + text3);  
//  
//      Thread.sleep(400);  
//  
//      driver.navigate().back();  
//  
//      Thread.sleep(400);
```

```
WebElement title4 = driver.findElement(Errorcode404);
```

```
title4.click();
```

```
String text4 = driver.getTitle();
```

```
System.out.println("Error status is" + text4);
```

```
Thread.sleep(400);
```

```
driver.navigate().back();
```

```
Thread.sleep(400);
```

```
// * WebElement title5 = driver.findElement(Errorcode408);
```

```
//
```

```
// title5.click();
```

```
//
```

```
// ts = (TakesScreenshot) driver;
```

```
//
```

```
// File src1 = ts.getScreenshotAs(OutputType.FILE);
```

```
//
```

```
// File des1 = new File("Screenshot/Error408.png");
```

```
//
```

```
// FileUtils.copyFile(src1, des1);
```

```
//
```

```
// Thread.sleep(400);
```

```
//
```

```
// driver.navigate().back();
```

```
//
```

```
// Thread.sleep(400);
```

```
WebElement title6 = driver.findElement(Errorcode500);
```

```
title6.click();
```

```
String text6 = driver.getTitle();
```

```
System.out.println("Error status is" + text6);
```

```
Thread.sleep(400);
```

```
driver.navigate().back();
```

```
Thread.sleep(400);
```

```
WebElement title7 = driver.findElement(Errorcode502);
```

```
title7.click();
```

```
String text7 = driver.getTitle();
```

```
System.out.println("Error status is" + text7);
```

```
Thread.sleep(400);
```

```
driver.navigate().back();
```

```
Thread.sleep(400);
```

```
//
```

```
// WebElement title8 = driver.findElement(Errorcode503);
```



```

//
//      title8.click();
//
//      ts = (TakesScreenshot) driver;
//
//      File src2 = ts.getScreenshotAs(OutputType.FILE);
//
//      File des2 = new File("Screenshot/Error503.png");
//
//      FileUtils.copyFile(src2, des2);
//
//      Thread.sleep(400);
//
//      driver.navigate().back();
//
//      Thread.sleep(400);
//
    }

}

```

ExtentManager Class

```

package Utilities;

import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.reporter.ExtentSparkReporter;

```

```

public class ExtentManager {
    private static ExtentReports extent;
    public static ExtentReports getInstance() {
        if (extent == null) {
            ExtentSparkReporter spark = new ExtentSparkReporter(
"Reports/ExtentReport.html");
            extent = new ExtentReports();
            extent.attachReporter(spark);
            extent.setSystemInfo( "Tester","BalaTeja");
            extent.setSystemInfo( "Project","Automation Test Practice");
        }
        return extent;
    }
}

```

TestListner Class

```

package Utilities;
import org.testng.ITestContext;
import org.testng.ITestListener;
import org.testng.ITestResult;
import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.ExtentTest;
public class TestListener implements ITestListener {
    ExtentReports extent = ExtentManager.getInstance();
    ExtentTest test;

```

@Override

```
public void onStart( ITestResult result) {  
    test = extent.createTest( result.getName());  
}
```

@Override

```
public void onSuccess(ITestResult result) {  
    test.pass("Test Passed Successfully");  
}
```

@Override

```
public void onFailure( ITestResult result) {  
    test.fail("Test Failed");  
    test.fail(result.getThrowable());  
}
```

@Override

```
public void onTestSkipped( ITestResult result) {  
    test.skip("Test Skipped");  
}
```

@Override

```
public void onFinish( ITestContext context) {  
    extent.flush();  
}  
}
```

Listner Class

```
package Utilities;

import org.testng.ITestContext;
import org.testng.ITestListener;
import org.testng.ITestResult;
import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.ExtentTest;

public class TestListener implements ITestListener {

    ExtentReports extent = ExtentManager.getInstance();

    ExtentTest test;

    @Override

    public void onTestStart( ITestResult result) {

        test = extent.createTest( result.getName());

    }

    @Override

    public void onTestSuccess(ITestResult result) {

        test.pass("Test Passed Successfully");

    }

    @Override

    public void onTestFailure( ITestResult result) {

        test.fail("Test Failed");

        test.fail(result.getThrowable());

    }

    @Override

    public void onTestSkipped( ITestResult result) {
```

```

        test.skip("Test Skipped");
    }

    @Override
    public void onFinish( ITestContext context) {
        extent.flush();
    }
}

```

POM.xml File

```

package Utilities;

import org.testng.ITestContext;
import org.testng.ITestListener;
import org.testng.ITestResult;
import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.ExtentTest;

public class TestListener implements ITestListener {

    ExtentReports extent = ExtentManager.getInstance();

    ExtentTest test;

    @Override
    public void onTestStart( ITestResult result) {
        test = extent.createTest( result.getName());
    }

    @Override
    public void onTestSuccess(ITestResult result) {
        test.pass("Test Passed Successfully");
    }
}

```

@Override

```
public void onTestFailure( ITestResult result) {  
    test.fail("Test Failed");  
    test.fail(result.getThrowable());  
}
```

@Override

```
public void onTestSkipped( ITestResult result) {  
    test.skip("Test Skipped");  
}
```

@Override

```
public void onFinish( ITestContext context) {  
    extent.flush();  
}  
}
```

Testing File

```
<suite name="Advanced Practice Suite Validation">  
  <test name="Functional UI Tests">  
    <classes>  
      <class name="com.test.automation.BaseTest"/>  
    </classes>  
  </test>  
</suite>
```

Output:

Default test

Tests passed/Failed/Skipped:	16/0/0
Started on:	Mon Jun 15 20:47:31 IST 2026
Total time:	157 seconds (157950 ms)
Included groups:	
Excluded groups:	